Literature Review: Three Threads Toward Conjecture 2.7

Tingley, Tight Partitions, and Quasisymmetric Functions

Robin Langer

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Abstract

We survey three approaches to Warnaar's Conjecture 2.7 (positivity of $Q_{n,c}(q)$ for cylindric plane partitions) and assess what each contributes and where each hits its limit. The three threads are: (1) Tingley's crystal base interpretation of cylindric partitions, (2) Kanade–Russell's tight cylindric partitions and functional equations, and (3) our own quasisymmetric function refinement and L-positivity results. All three illuminate different facets of the conjecture but each encounters the same obstacle: the $[z^n]$ extraction that isolates CPPs with a fixed maximum entry.

1 Thread 1: Tingley — Crystal bases and cylindric partitions

1.1 Main result

Tingley [**?**] proved that cylindric plane partitions are crystal basis elements of $\widehat{\mathfrak{sl}}_n$. He establishes three equivalent combinatorial models for the crystal graph:

1. **Hayashi realization**: partitions with crystal operators e_i, f_i acting by adding/removing boxes colored by residue mod n.
2. **Abacus model**: beads on an n-strand abacus, with crystal operators sliding beads between strands.
3. **Kyoto path model**: semi-infinite tensor products of elements from a perfect Kirillov–Reshetikhin crystal $B_{r,s}$.

Theorem 1 (Tingley, Theorem 3.14)**.** *The colored directed graph formed by tight, descending abacus configurations is a realization of the crystal graph for the $\widehat{\mathfrak{sl}}_n$ representation $V(\Lambda)$.*

The central application: Tingley gives a **representation-theoretic proof of Borodin's product formula** [**?**]. The generating function $\mathrm{GK}_c(1,q)$ is identified with the principally specialized character of an integrable highest-weight $\widehat{\mathfrak{sl}}_n$-module, and characters of integrable modules factor by the Weyl–Kac character formula.

1.2 The underlying chain

The connection runs through the corner transfer matrix (CTM) method of Baxter. In a solvable lattice model satisfying the Yang–Baxter equations, the CTM eigenstates at $q = 0$ form the crystal basis $B(\Lambda)$. A *path* through this crystal is a sequence $(b_1, b_2, \ldots, b_L) \in B_{r,s}^{\otimes L}$. When the path is periodic (ground-state boundary condition), it becomes a cylindric partition. The energy function $H : B_1 \times B_2 \to \mathbb{Z}$ tracks the q-weight:

$$X(B, \Lambda; q) = \sum_{\text{paths } p} q^{\text{energy}(p)} = \mathrm{GK}_c(1, q).$$

This one-dimensional configuration sum equals the CPP generating function, which factors into Borodin's product.

1.3 The bijection chain

There is a chain of weight-preserving bijections:

Cylindric partitionsTingleyCrystal pathsKyotoSemi-infinite tensor productsOSSRigged configurations

where OSS refers to Okado–Schilling–Shimozono [**?**]. Each arrow preserves the generating function weight (energy $\leftrightarrow$ size $\leftrightarrow$ cocharge).

1.4 Relevance to Conjecture 2.7

If $Q_{n,c}(q)$ could be interpreted as a weight multiplicity or branching coefficient inside the crystal, positivity would be automatic—crystals count things. The obstacle is that the $[z^n]$ extraction (fixing the maximum entry of a CPP) does not have a known crystal interpretation. The maximum entry $n(\pi) = \max\{\nu_j^{(i)}\}$ is not a crystal-theoretic statistic: it depends on the absolute values of entries, not on their relative positions or the action of the operators e_i, f_i.

Remark 2. *Tingley's crystal structure lives on the* full *set of CPPs (all max entries). Restricting to* $\max = n$ *breaks the crystal into fragments that do not form sub-crystals in general. This parallels the failure of quasi-symmetry at fixed max entry (Thread 3).*

1.5 Key references

- Tingley [**?**]: the central bridge paper.
- Kang–Kashiwara–Misra [**?**]: perfect crystals and the path realization theorem.
- Okado–Schilling–Shimozono [**?**]: path-to-rigged-configuration bijection.
- Nakayashiki–Yamada [**?**]: Kostka polynomials and energy functions in solvable lattice models.

2 Thread 2: Kanade–Russell — Tight cylindric partitions

2.1 Main result

Kanade and Russell [**?**] define *tight* cylindric partitions, where consecutive stack heights differ by at most 1 (a stricter condition than ordinary interlacing). They prove:

Theorem 3 (Kanade–Russell, Theorem 4)**.** *For a profile* $c = (c_1, \dots, c_r)$ *with* $m = c_1 + \cdots + c_r$*, the univariate generating function for tight CPPs is*

$$T_c(1,q) = \Pr\big(L_{\widehat{\mathfrak{sl}}_n[r]}(\Lambda(c))\big) = \frac{(q^r;q^r)_\infty\,(q^m;q^m)_\infty^{r-1}}{(q)_\infty^r} \prod_{1\le i<j\le r} \theta\big(q^{j-i+c_i+\cdots+c_{j-1}};\, q^m\big).$$

The proof uses **Corteel–Welsh functional equations**: a system of equations for the bivariate generating function $T_c(z,q)$, solved via inclusion-exclusion and induction on z-exponents. This is a purely analytic approach—no representation theory, no crystal bases.

2.2 Additional results

- **Theorem 10**: Corteel–Welsh functional equation analogue for r-rowed tight CPPs.

- **Theorem 13**: Bivariate closed forms for 2-rowed tight CPPs. For profile (c_1, c_2), the generating functions reduce to $(-q;q)_\infty$ times Gordon–Andrews or Andrews–Bressoud identities.
- **Theorem 14**: Diamond relations reformulation.
- The 3-row case remains open (only conjectural bivariate formulas for small profiles).

2.3 Relevance to Conjecture 2.7

The Corteel–Welsh functional equations are powerful for computing generating functions and proving product formulas, but they do *not* directly address positivity. The explicit double-sum formulas for $[y^n]G_c(y,q)$ (Theorem 3.2 of Corteel–Welsh [**?**]) involve terms with alternating signs for profiles like $(3,0,1)$ and $(2,2,0)$. Positivity of $Q_{n,c}(q)$ requires delicate cancellation that the functional equations cannot explain.

The functional equations are a computational oracle, not a conceptual explanation. They tell us *what* $Q_{n,c}(q)$ equals, but not *why* its coefficients should be positive.

Remark 4. *The tight CPP product formula is structurally parallel to Borodin's identity (both are principally specialized $\widehat{\mathfrak{sl}}_n[r]$ characters). This suggests that an alternative proof via crystal bases (extending Tingley's approach to tight CPPs) should be possible and would deepen the representation-theoretic picture. It could also reveal whether analogous Q-polynomials exist for the tight case.*

3 Thread 3: Quasisymmetric functions and L-positivity

3.1 Main results

In [**?**], we showed that the CPP frequency generating function

$$\widetilde{\mathrm{GK}}_c(\mathbf{z}) = \sum_{\pi \in \mathrm{CPP}(c)} z_1^{\#_1(\pi)}\, z_2^{\#_2(\pi)} \cdots$$

(where $\#_k(\pi)$ counts parts equal to k) lies in the ring QSym of quasisymmetric functions. This follows from Stanley's P-partition theory: the interlacing conditions depend only on the relative order of entries, not their absolute values.

Observation 5 (Verified: 55 profiles, $d \leq 5$)**.** *$\widetilde{\mathrm{GK}}_c$ is L-positive: $\widetilde{\mathrm{GK}}_c = \sum_\alpha b_\alpha(c)\, L_\alpha$ with $b_\alpha \in \mathbb{Z}_{\geq 0}$.*

The stabilized L-coefficients follow $b^{\ell(\alpha)}$ patterns ($b = 2$ for $d = 2$, $b = 5$ for $d = 4$), suggesting a product structure. By Tewari–van Willigenburg [**?**], L-positivity is equivalent to the existence of a 0-Hecke module whose character is $\widetilde{\mathrm{GK}}_c$.

Conjecture 6 (Descent statistic)**.** *There exists a natural descent-type statistic on cylindric partitions such that $\widetilde{\mathrm{GK}}_c = \sum_\pi L_{\alpha(\pi)}$, where $\alpha(\pi)$ is the descent composition of π.*

3.2 The dead end

L-positivity does **not** extend to the Q-polynomials. Two obstacles:

1. **Max-entry constraint breaks quasi-symmetry.** The individual slices $\tilde{g}_n = \sum_{\max(\pi)=n} z_1^{\#_1} \cdots$ are *not* quasi-symmetric: fixing max $= n$ privileges the value n, breaking the relabelling argument.

2. **Cross-frequency cancellation.** The alternating-sign factor $(zq;q)_\infty$ introduces cancellation that crosses between different frequency vectors. Computationally, $\sim$3–6% of the cancellation in Q_n is cross-frequency: the net coefficient at individual frequency vectors is negative, and only the aggregation across all compositions of the same weight (under $z_k \mapsto q^k$) yields the positive Q_n coefficient.

The positivity of $Q_{n,c}(q)$ is an **intrinsically one-variable phenomenon**: the specialization $z_k \mapsto q^k$ is essential and cannot be avoided within QSym.

3.3 Connection to Lam–Pylyavskyy Conjecture 5.5

Conjecture 5.5 of [**?**] characterizes when products $L_\alpha \cdot L_\beta$ are maximal in the L-positivity order (iff the pair $\{\alpha, \beta\}$ is stable under cell transfer). This is relevant because the stabilization pattern of $\widetilde{\mathrm{GK}}_c$ suggests a product structure in the L-basis, and Conjecture 5.5 would constrain its form. However, this enriches the understanding of $\widetilde{\mathrm{GK}}_c$ itself (the unconstrained generating function) without bridging the gap to $Q_{n,c}$ positivity.

4 Synthesis

The three threads illuminate different facets of the same problem:

Thread	**What it gives**	**Where it stops**
Tingley (crystals)	*Why* Borodin's product holds (characters of $\widehat{\mathfrak{sl}}_n$ modules)	$[z^n]$ extraction has no crystal interpretation
Kanade–Russell (functional eqns)	*What* $G_c(y,q)$ equals (explicit double sums)	Formulas have mixed signs; no positivity mechanism
QSym / L-positivity	*Structure* of $\widetilde{\mathrm{GK}}_c$ (L-positive, 0-Hecke module)	Positivity is one-variable; cross-frequency cancellation

All three approaches converge on the same obstacle: the $[z^n]$ extraction that isolates CPPs with a specific maximum entry. In the crystal picture, fixing max entry breaks the crystal into non-sub-crystal fragments. In the QSym picture, it breaks quasi-symmetry. In the functional equation picture, it introduces alternating signs.

4.1 What would a proof need?

A proof of Conjecture 2.7 likely requires one of:

1. A **combinatorial interpretation** of $Q_{n,c}(q)$ as a counting function—finding what $Q_{n,c}$ counts.

2. A **crystal-theoretic interpretation** of the $[z^n]$ extraction—expressing $Q_{n,c}$ as a weight multiplicity or branching coefficient.

3. A **new positivity mechanism** that works in one variable—perhaps a positivity-preserving transformation (in the spirit of the Borwein program) or a connection to Kazhdan–Lusztig theory.

The most promising direction may be (2): if the "max = n slice" of the crystal corresponds to some natural representation-theoretic object (a Demazure module? a level restriction? a truncated path space?), then positivity would follow from the general theory. This is the direction suggested by the parallel between Borodin's identity and the Kanade–Russell product formula—both are $\widehat{\mathfrak{sl}}_n[r]$ characters, and the extraction of Q_n should have a meaning in that world.

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